

Patterns of Secondary Palatalisation in Tiv Nouns.

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Abstract

This study examines secondary palatalisation patterns in Tiv nouns. The objectives were to determine the physiological basis of the patterns, describe the environments where these patterns occur, and explain their effects on Tiv grammar. Data were gathered from the field through participant observation, consultations, non-directive interviews, and relevant scholarly materials. The focus of the study is on the Gboko dialect. A purposive sampling technique was employed. A total of 85 monolingual Tiv native speakers were selected as respondents, including 3 male and 2 female adults, aged 40-70, each from one of the 17 council wards within the local government. The data were analysed using Articulatory Phonological (AP) theory. The findings revealed that, besides being morphologically and phonologically conditioned, secondary palatalisation in Tiv nouns is based on the presence of the front high unrounded /i/ as a V-shaped syllable in the physiological configuration of the noun, serving as a pre-palatalisation cue; the presence of /j/, which functions as a palatalisation trigger immediately after the onset of the second syllable; and changes in the constriction location (CL) and constriction degree (CD) of the target consonants. It was further found that all target consonants are palatalised before the palatal glide /j/. Secondary palatalisation in Tiv nouns is restricted to the onset of the second syllable, in line with Tiv phonotactics. The failure to account for the primary and secondary places of articulation of the target consonants results in ungrammatical realisations in usage contexts. Based on the findings, it is recommended that further research be conducted to explore other forms and patterns of palatalisation in Tiv.

Keywords: palatalisation, gestural coordination, articulators, constraints, constriction

Introduction

The Tiv language is one of the major languages spoken in central Nigeria. It belongs to the Benue-Congo subgroup of the Bantu language family. It has over six million speakers across five Nigerian states. Tiv employs a morpheme-based morphological process where morphemes are concatenated for derivational (both syntactic and morphological) and inflectional purposes. These morpheme concatenations often lead to various sound changes with different grammatical effects. However, less attention has been given to the sound changes that occur in Tiv due to morphological and phonological interactions. Most previous studies on the Tiv language focus on morphology, phonology, syntax, and semantics without considering the interrelationship between these aspects of linguistic analysis. Consequently, there is a notable lack of literature on the sound changes resulting from morphology-phonology interactions and their implications for Tiv grammar and phonology. It is within this context that this study aims to investigate the morphologically and phonologically conditioned patterns of secondary palatalisation in Tiv nouns.

Palatalisation, according to Batemen (2007, p. 6), “is any instance where a consonant changes its place features to palatal-like, regardless of the nature of the trigger; it is any instance of a consonant acquiring a secondary palatal articulation”. He explains that palatalisation is a term for the articulation involving a movement of the tongue toward the hard palate; it is the interaction of some consonants and vowels in a language, which is constrained by language-specific phonotactics. According to this author, palatalisation takes place at the region behind the alveolar ridge through the hard palate to the soft palate. This area, according to him, covers the alveo-palatal and palatal places of articulation.

Bhat (1978) holds that palatalisation takes place if there is any shift in the primary place of articulation of a sound towards the palatal region. The author argues further that the shift could be from the front or from the back, converging within the alveo-palatal, hard palate and soft palate. Bhat’s definition of palatalization covers cases that meet one or both of the following conditions (1) The environment that induces the change must be a “palatalizing environment” (i.e. it must be a front vowel, a palatal semivowel, or a palatal or palatalized consonant) (ii) The sound that results from the change must be palatal or must have a secondary palatal articulation. Tongue-raising (essentially secondary

palatalisation) occurs more with apical (sounds articulated with the tip of the tongue) and labial sounds and is triggered by a following high (particularly front) vowel or semivowel. Tongue-fronting occurs more with velars and is triggered by a front vowel (not necessarily high). Spirantisation may occur along with the palatal glide and the trill /r/ among others (Bhat 1978:56), and it may occur in combination with tongue-fronting and tongue-raising. Velars may be affected by fronting, raising and spirantisation at the same time, resulting in /k/ => /tʃ/ (Bhat 1978, pp.49-51).

Ekpe (2013) established a distinction between the palatal sounds and the palatalised sounds. According to him, palatal sounds are mainly articulated in the palatal region, whereas sounds produced elsewhere but with palatal features are considered palatalised. That means palatalisation cannot affect palatal sounds because they are inherently palatalised. The physiological make-up of palatal sounds makes it impossible for them to be palatalised because the same articulators cannot possibly maintain varying degrees of stricture simultaneously. The sounds that are palatalised are non-palatal sounds.

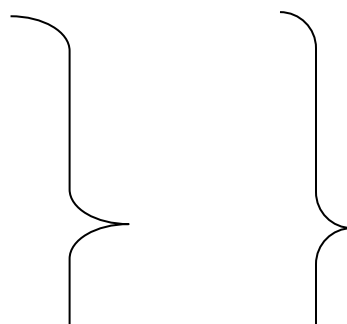
The above definitions help this study in no small measure in understanding the physiological make of palatalisation, the difference between palatal and palatalised sounds and the constriction location of palatalised sounds. They also provide the study with the clue that palatalisation is universal, but patterns of occurrence are language-specific. All these provide a challenge for the investigation of patterns of palatalisation in Tiv and thus the reason for the study of patterns of secondary palatalisation in Tiv nouns.

Palatalisation targets

Palatalisation targets are consonants. (Bateman, 2007) says the target consonants are considered in terms of place of articulation and sometimes manner of articulation. Chen (1973) opinionated that most of the palatalisation targets are obstruents, with stops out numbering fricatives and affricates. The next best targets are the nasals, followed by laterals, and finally by rhotics.

Extrapolating from Chen (1973) and Bateman (2007), this study considers as palatalisation targets in Tiv:

[p] : voiceless bilabial stop
[b] : voiced bilabial stop



[t] : voiceless alveolar stop
 [d] : voiced alveolar stop
 [k] : voiceless velar stop
 [g] : voiced velar stop
 kp]: voiceless labio-velar stop oral stop sounds , abstruents
 gb]: voiced Labio-velar

[f] : voiceless labiodental fricative	}	fricatives		
[v] : voiced labiodental fricative				
[h] : voiceless glottal fricative				
[m] : bilabial nasal stop	}	nasal stops	}	sonorants
[mb]: pre-nasalised bilabial stop				
[n] : voiced alveolar nasal				
[ŋ] : (voiced) velar nasal				

This consideration is based on the constriction degree of the target consonants
 (Source: field work January 2023)

The study groups the target consonants based on constriction locations as follows

1. **LABIAL:** [+labial]. These consonants are produced with the lips. These include:
 - i. Bilabials are produced with the two lips touching. /p/, /b/, /m/, /mb/
 - ii. Labiodental: Produced with the lips touching the front teeth. /f/, /v/
 - iii. Labio-velar: lip closed with the body of the tongue touching the velum. /kp/, /gb/
2. **CORONAL:** These consonants are produced using the tip of the tongue. The coronal consonants in Tiv are the following: **Alveolars** /t/, /d/, /l/, produced with the tip or blade of the tongue touching the alveolar ridge
3. **DORSAL:** Consonants are produced with the body of the tongue. They include:

- i. **Palatal:** Produced with the body of the tongue touching the (hard) palate. /j/ [palatal, narrow] stiffer
- ii. **Velar:** Produced with the body of the tongue touching the velum (soft palate). /k/, /g/, /ŋ/.

4. GLOTTAL: A consonant is produced with the epiglottis. The tongue is not active, and there is no more specific place of articulation for these sounds. [h] has the feature [+glottal].

(source: Ayagah 2022).

Palatalisation triggers

The palatalised configuration is also composed of a ‘trigger’. The trigger is always a vowel or semi-vowel, while the target is a consonant (Bhat, 1978). Chen (1973), Bhat (1978), Hall (2000) and Bateman (2007) specify the typical palatalisation triggers as the front vowels i and e, and the palatal glide j. Triggers are universal, but their combination is language-specific and is not random but within predictable patterns according to vowel height and backness. Bateman (2007, p.63) explains this with instances as follows:

There are languages where i is the only palatalization trigger (e.g. Apalai, Basque, Carib, Fongbe, Kayardild), languages in which only the palatal glide is a trigger (e.g. English, Shilluk, Yagua, Zoque) and languages where i and the glide j are triggers (e.g. Luvale, Nishnaabemwin, Yimas). There are many languages where only front vowels are the triggers (e.g. Dakota, ɛ, y, ø and the diphthong ea triggers, primarily including i and e but also Hausa, Mwera, Nupe, Romanian, Turkish), and a few others in which front vowels and high vowels which are further back (primarily u, but also ʊ and ɨ) trigger palatalization, in addition to the usual i (e.g. Coatzacoapan Mixtec, Maori, Sentani, Tohono O’Odham)

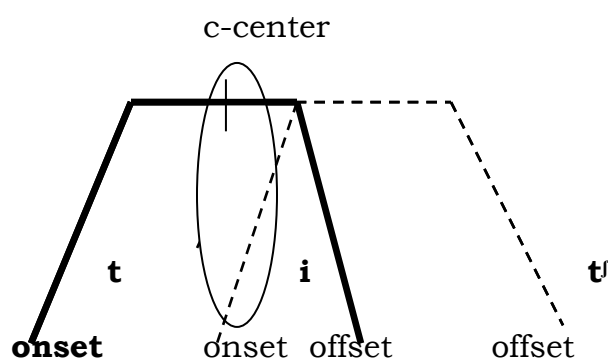
The trigger vowel in palatalisation is determined by the height and frontness of the tongue, while the target consonant is considered in terms of place of and sometimes manner of articulation. Bateman (2007:272) says that the “implicational hierarchy of palatalisation triggers is explained in terms of the gestural properties of the sound involved in palatalisation”. In Tiv palatalisation, the trigger is the palatal glide /j/.

Types of palatalisation

Bateman (2007:2) distinguishes two types of palatalisation: full and secondary palatalisation. In full palatalisation, the consonant shifts its primary place and often its manner of articulation while moving toward the palatal region of the vocal tract. Secondary palatalisation, on the other hand, involves the co-articulation of the target consonant with a following palatal off-glide. This classification is based on the degree of temporal overlap of gestures (Bateman, 2007), as shown in 1 and 2 below

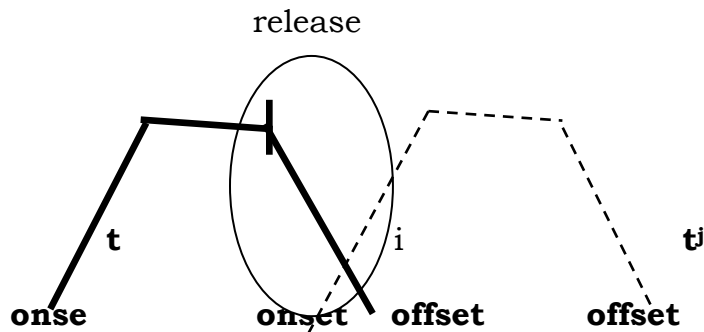
1. Full palatalisation

In full palatalisation, the onset phase of the vocalic gesture (dotted line) is synchronised with the c-centre phase of the consonantal gesture (solid line), representing a high degree of temporal overlap. The c-centre is a general landmark, the key idea being that the onset of the vocalic gesture is coordinated with some point during the closure phase of the consonantal gesture, which is within the c-centre landmark phase



On the other hand, in secondary palatalisation the onset phase of the vocalic gesture (dotted line) is bonded with the release phase of the consonantal gesture (solid line), as in (2). This represents a minimal degree of temporal overlap, where the consonant is fully articulated and also has a secondary tongue body palatal articulation.

2. Secondary palatalisation



Source: Bateman (2008, p.29)

In addition to the above two types of palatalisation, Bateman (2007:10) distinguishes between purely phonological and morpho-phonological contexts of palatalisation. According to Bateman (2007), purely phonological palatalisation occurs across the board in a language, which is understood as allophony. Some examples of purely phonological palatalisation cited by Bateman (2007:11) are given below:

- (i) /pitiko/ → [pitʃiko] 'small' (Apalai, Carib)
- (ii) /kukira/ → [kutʃiæ] 'exceed' (Nkore-Kiga, Bantu)
- /egjo/ → [edʒo] 'that'
- (iii) /gɛbam/ → [gʲɛ bam] 'pound' (Koromfe, Gur)
- /giram/ → [gʲiram] 'judge'

A morpho-phonological context is one where palatalisation is restricted to certain morphological forms, but where it is phonologically conditioned by the presence of a common palatalising trigger (Bateman, 2007:11). The major types of morpho-phonological palatalisation are given below:

a. Triggered by affixation: affix contains a palatalising trigger.

- (i) /ka za / 'hen' → [kadʒi] 'hens' (Hausa, Chadic)
- /mota/ 'car' → [mototʃi] 'cars'
- (ii) /fak/ 'make/do' (1Sg 3Pl) → [fatʃe] 'make/do' (3Sg) (Standard Romanian).

b. Either triggered by affixation with an opaque trigger, or no apparent trigger but expressing a morphological property:

- (i) /fak-i/ → fatʃ 'make/do' (2 Sg) (Standard Romanian)
- (ii) /brad-i/ → [brazj] 'fir tree'

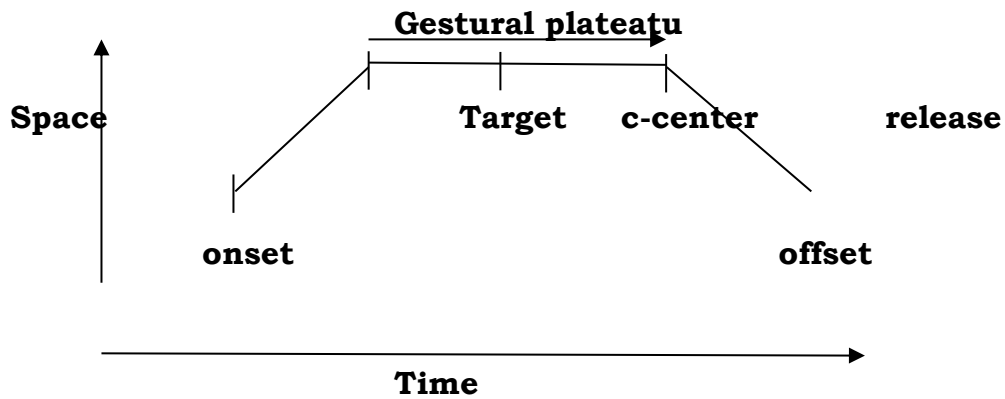
From data collected from the field vis-à-vis the reviewed related literature in this study, it will be right to say that palatalisation typology in Tiv nouns is secondary palatalisation in that the target consonants do not shift completely their primary place and manner of articulation toward the palatal region of the vocal tract rather, they are co-articulated with a following palatal off glide as in i-tyar [itʲa:]

Theoretical framework

This work adopts the Articulatory model of generative phonology propounded by Browman & Goldstein (1990). In articulatory phonology, the articulatory gesture is viewed as the hub of phonological description. Bateman (2007) refers to gesture as a dynamically defined, spatio-temporal unit. Gestures consist of the formation of a constriction by some articulators at some places in the vocal tract. The vocal tract contains some variables which specify the spatial goals of that constriction. This specification consists of the ‘articulator set’ employed in producing the constriction, the constriction location (CL), and the constriction degree (CD) (Gufos 2001).

Gestures contrast on the basis of the different constrictions in the vocal tract. The CD variable, for example, takes on a range of values with five categorical distinctions: [closed], [critical], [narrow], [mid], and [wide], while CL specifies the place of the constriction in the vocal tract. It takes the values of [labial], [dental], [alveolar], [postalveolar], [palatal], [velar], [uvular], and [pharyngeal], based on Ladefoged (1989).

Gestures also have a temporal dimension. A gesture unfolds in time. There are a few key points that can be identified during the movement of a gesture. First, there is the onset of movement - the point in time when the articulator begins to move towards its specified target. Second, as the articulator progressively approaches its target, at some point it attains it. This point in time is the achievement of target (Gafos, 2001). What this means is that when the constriction is held for some time, the articulator releases it and starts its movement away from the target - the point in time known as the articulatory release. These points, Gafos (2001, p.7) refers to them as “landmarks” and diagrammatically illustrate them as shown in Figure 1:



Source: Gafos (2001, p.7)

Figure 1: Landmark in gestural life

It can be deduced from Figure 1 that the curved path is similar to the continuous movement of an articulator, such as the tongue tip-blade, in space (vertical axis) and time (horizontal axis). The portion of the curve between target and release is called the '**gestural plateau**'. The gestural plateau, as explained by Bateman (2007), is:

The time during which the constriction is held by a particular articulator, such as the tongue tip or the tongue body, between the target and the release, and the c-centre (constriction phase centre) is the mid-point of the gestural plateau. Thus, for an [m] gesture, the onset occurs when the lip articulators begin the movement toward the constriction location, the target is reached during lip closure; the c-centre temporal landmark is reached while the lips are closed. Lip opening marks the release of the gesture, and the release offset occurs when the lips cease to be under active control

It can be extrapolated from the foregoing review that gestures are abstract phonological units with an internal spatio-temporal organisation, and they make direct reference to physical implementation by the articulators. They are both phonological representation and phonetic implementation.

Gestural coordination

Gestural coordination is a basic principle of AP, which is the basis for phonological processes. As speech unfolds in real time, gestures coordinate with

one another to achieve the speech output, as they transition from one to the other. In other words, some phase (time point) of one gesture must be synchronised with some phase of another adjacent gesture (Bateman 2007).

Gafos (2002, p.9) defines a gestural coordination relation as “a relation between two gestures stating that a specified landmark of one gesture is synchronous with a specified landmark of another gesture” (Gafos 2002:9). Browman and Goldstein (1986) maintain that gestural coordination can result in: (i) gestural magnitude reduction in space and/or time, and (ii), in gestural temporal overlap.

It can be deduced from the foregoing that in articulatory phonology, the articulatory gesture is viewed as the hub of phonological description. A gesture is identified in terms of an articulator (e.g. lips, tongue), a constriction location (place of articulation) and a constriction degree (manner of articulation). In addition, gestures have duration and are timed with respect to each other. As speech occurs in real time, gestures must coordinate with one other and as a result they can temporally overlap (Bateman, 2007).

The review of Articulatory phonological theory indicates that it specifies the roles of oral articulators in the analysis of human speech sounds and is therefore apt for the analysis of patterns of secondary palatalisation in Tiv. This is in consideration of the fact that palatalisation arises as a result of the natural interaction of the oral articulators during speech and the consequences of that interaction on speech production. The degree of gestural overlap and the participating oral articulators determine the type of palatalisation that results from gestural coordination. This study adopts Bateman’s (2007) submission that articulation is the principal reason for palatalisation.

Methodology

This study adopted a qualitative research design. The study obtained data from the field from adult/elderly monolingual native speakers of Tiv language, the clergy, the academics and members of Tiv Language Studies and Development Association (TLSDA) through participant observation, consultations, interviews and the researcher’s intuitive knowledge. A total of 75 respondents were selected, 40 respondents were representatives of the whole

population of native Tiv speakers who resided in the rural areas, 8 respondents from the church, 12 respondents from academia and 15 respondents were members of the Tiv Language Studies and Development Association (TLSDA). The clergy, the academics and members of Tiv Language Studies and Development Association (TLSDA) were selected as representatives of people specialised in formal knowledge of Tiv language. The reason for selecting these people was to use their formal knowledge to filter the data collected from the monolingual speakers, who do not have formal knowledge of Tiv language.

Articulatory phonology and secondary palatalisation patterns in Tiv

nouns.

Secondary Palatalisation in Tiv nouns is both morphologically and phonologically conditioned phonological process. It is morphologically when it is determined by affixation. Morphologically induced secondary palatalisation in Tiv noun occurs in environments where the front high unrounded vowel /i/ is prefixed to the nominal base whose initial consonants is any of the following /p/, /b/, /t/, /d/, /k/, /kp/, /g/, /gb/, /f/, /v/, /m/, /mb/, /ŋ/, /h/ and /l/. The prefixation of /i/ to the nominal base further conditions the insertion of /j/ immediately after the target consonants in the second syllable. The consequence of the insertion of the /j/ is a change in the constriction location (CL) and constriction degree (CD) of the target consonants, as the tongue moves to the palatal region. This process is shown in the example

Example 1

Singular	IPA	Gloss	Plural	IPA	Gloss
tar	[ta:]	nation	i-tyar	[i.tja:]	nations
kwa	[k ^w a]	leaf	i-kya	[i.kja]	leaves
bua	[bua]	cow	i-byua	[i.bjoa]	cows
takeda	[takeda]	book	i-tyakeda	[i.tjakeda]	books
tom	[tom]	job	i-tyom	[i.tjom]	jobs

There are instances in Tiv nouns where the physiological make up of a noun contains the front high unrounded vowel /i/ as initial syllable, accompanied by

a syllable whose onset is any of the following consonants /p/, /b/, /t/, /d/, /k/, /kp/, /g/, /gb/, /f/, /v/, /m/, /mb/, /ŋ/, /h/ and /l, and the palatal glide /j/ as shown in example (2) without affixation. This usually takes place with singular nouns as shown on example 2

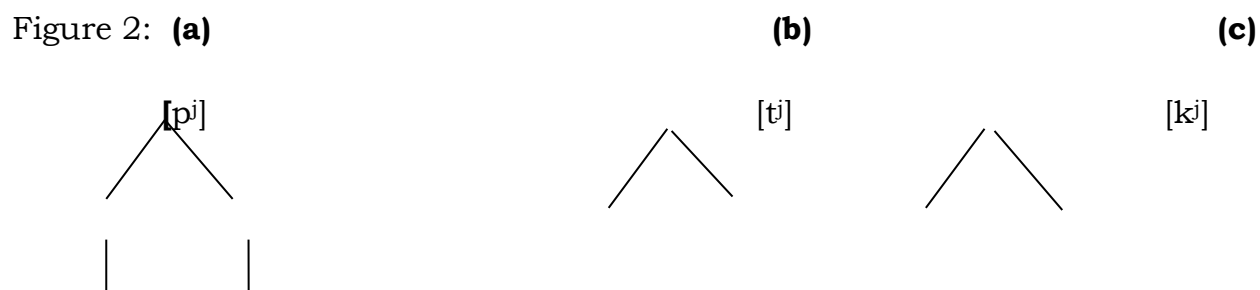
Example 2

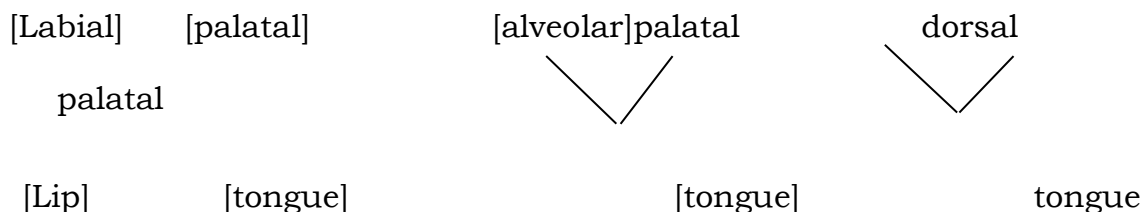
i. ifyam	[iɸam]	frog
ii. ivyaan	[ivja:n]	arrow
iii. ihyongo	[ihjonŋo]	pool of water
iv .igywa	[igjwa]	big gun
v. imyôngo	[imjonŋo]	together

It is therefore right to say that secondary palatalisation in Tiv nouns is morphologically conditioned in its plural formation and phonologically determined in its singular noun configurations. Generally speaking, secondary palatalisation in Tiv nouns is based on three conditions:

1. The presence of the front high unrounded /i/ as a v-shaped syllable in the physiological configuration of the noun as a pre-palatalisation clue
2. The presence of /j/ functions as a palatalisation trigger immediately after the onset of the second syllable.
3. A change in the constriction location (CL) and constriction degree (CD) of the target consonants.

As earlier mentioned, Articulatory Phonology provides explanations of why palatalisation happens in languages, Tiv. It indicates that palatalisation is a consequence of the natural interaction of the oral articulators during speech. Drawing from the provisions of articulatory phonology, this work examines palatalisation as a natural result of coordinating consonantal and vocalic gestures in the course of speech production, as illustrated in Figure 2





The analysis of the interaction of oral articulators in Figure (a) indicates that the tongue and the lips are separate articulators, free to move independently of one another. This means the tongue and the lips can have no overlap, overlap minimally, partially or fully. No palatalisation results when they do not overlap. When they minimally or partially overlap, the resultant phonological outcome is secondary palatalisation and full palatalisation results when the overlap is full. The lips and the tongue can each perform their task sequentially or simultaneously. There is no compulsion that the lips and tongue must blend. In the process of palatalisation, as indicated in Figure 2a, the lips gesture must perform the task of lip closure to achieve a [p], while that of the tongue gesture is of performing the task of achieving the palatalisation trigger [j]. The implication here is that if a lip gesture and the following tongue gesture were fully overlapped, the lip gesture would obscure the tongue gesture since the lip is physically in front of the tongue. The outcome of the complete gestural overlap of [p] and [j] would be [p]. There is, however, minimal temporal overlap, which synchronises the onset phase [j] with the release phase of [p], leading to [pʲ] acquiring a labial palatal articulation as illustrated in Figure 3

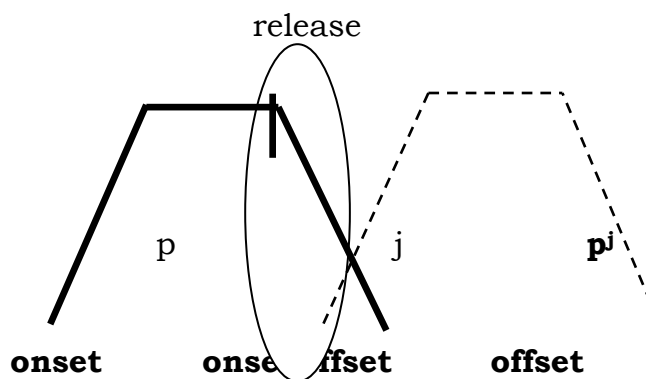


Figure 2b and c indicate that coronals and dorsals are products of the tongue. Coronals are articulated with the tongue tip, while dorsals are articulated with the back of the tongue – both of which are sub-articulators of the more basic

tongue articulator. These sub-articulators of the tongue are very closely connected, much more so the either is to the lips. This closeness implies that when the tongue body makes a gesture, it is likely for the movement to affect the movement of the tongue tips, and vice versa. This study draws inference from Bateman (2007, pp. 210-211) assertion that “coordination of consonants with a following high vocoid can result in full palatalisation only if the articulator [tongue] alone is employed; if the lips and the tongue are employed, only secondary palatalisation can ensure” to assert that Tiv does not show full palatalisation because of the dominance labial consonant targets.

It could be deduced from the foregoing that articulatory phonology makes a clear distinction between labial consonants on the one hand and coronal and dorsal consonants on the other. It has to be noted that traditional phonological theory treats all three places of articulation equally, without according to special status to one place versus another. But Bateman’s (2007) investigation on patterns of palatalisation indicates that labial consonants are and should be treated as special. The speciality of labials is considered the most basic distinction between them and other places of articulation. They are the only sounds produced with the lips. This study’s account of palatalisation, therefore, builds on Articulatory phonological theory, which captures this special status of labials and the tongue, the main articulator.

Secondary palatalisation in Tiv nouns manifests as a secondary articulation process where the feature of palatal glide /j/ is superimposed on an adjacent onset consonant. A secondary palatalised consonant is marked with the diacritic [j], hence /py/ is realised as [p^j], a palatalised voiceless bilabial plosive. The data in example (3) exemplifies the secondary palatalisation process of nouns in syllables whose onset is the labials

Example 3. palatalisation of labial consonants in Tiv nouns

$$(a). p \longrightarrow p^j / \left\{ \begin{array}{l} +i\text{-initia} \\ -\text{voice} \\ +\text{bilabial} \\ + \text{glide } /j/ \end{array} \right\} \text{ as in ipyaven, ipyusu, ipyareghyoloo}$$

$$/ \left\{ \begin{array}{l} +i\text{-initia} \end{array} \right\}$$

(b). b → bj $\left\{ \begin{array}{l} +\text{voiced} \\ +\text{bilabial} \\ +\text{glide } /j/ \end{array} \right\}$ as in ibyaven, ibya, ibyumeor.

(c). m → mj $\left\{ \begin{array}{l} +i\text{-initial} \\ +\text{bilabial} \\ +\text{nasal} \\ +\text{glide} \end{array} \right\}$ as in imyaagh, imyôngo, imyemegh

(d). kp → kpj $\left\{ \begin{array}{l} +i\text{-initial} \\ +\text{labial} \\ +\text{glide} \end{array} \right\}$ +velar as in ikpya, ikpye, ikpyagher

(e). gb → gbj $\left\{ \begin{array}{l} +i\text{-initial} \\ +\text{labial} \\ +\text{glide} \end{array} \right\}$ +velar as in igbya, igbyenger, igbyende

(f). mb → mbj $\left\{ \begin{array}{l} +i\text{-initial} \\ +\text{labial} \\ +\text{glide} \end{array} \right\}$ +pe-nasal as in igbya, igbyenger, igbyende

(g). f → fj $\left\{ \begin{array}{l} +i\text{-initial} \\ +\text{labial} \\ +\text{glide} \end{array} \right\}$ +dental as in ifyam, ifyer, ifyeesegeh

(h). v → vj $\left\{ \begin{array}{l} +i\text{-initial} \\ +\text{labial} \\ +\text{glide} \end{array} \right\}$ +dental as in ivyaan, ivyever, ivyaavyase, ivyeren

This study proposes that palatalisation of labials in Tiv nouns is secondary palatalisation. This proposal is based on the nature of the coordination of the gestures involved. The analysis of example 3 a-h portrays that the lips gesture and the following tongue gesture are fully overlapped. As earlier said, lips are physically in front of the tongue and would have obscured the tongue gesture if the overlap were full.

Example 4 : palatalisation coronal consonants

$\left\{ \begin{array}{l} +i\text{-initial} \end{array} \right\}$

- (a). t $\longrightarrow$ tʲ / $\left\{ \begin{array}{l} +\text{coronal} \\ +\text{alveolar} \\ +\text{glide} \end{array} \right\}$ as in ityav, ityenger, ityev
- (b). d $\longrightarrow$ dʲ / $\left\{ \begin{array}{l} +\text{i- initial} \\ +\text{coronal} \\ +\text{alveolar} \\ +\text{glide} \end{array} \right\}$ as in idya, idyeregeh, idyondon
- (c). l $\longrightarrow$ lʲ / $\left\{ \begin{array}{l} +\text{i- initial} \\ +\text{coronal} \\ +\text{alveolar} \\ +\text{glide} \end{array} \right\}$ as in ilyam, ilyegher, ilyôngo

The process palatalisation coronal consonants in Tiv nouns is characterized by tongue-raising triggered by the feature of palatal glide /j/. Bhat (1978) calls this process ‘tongue-raising’, while Bateman (2007), Hall (2000), and Kochetov (2011) all call it ‘secondary palatalization’ as shown in example 4

Example 5. Palatalisation dorsal consonants

- (a). k $\longrightarrow$ kʲ / $\left\{ \begin{array}{l} +\text{i- initial} \\ +\text{dorsal} \\ +\text{velar} \\ +\text{glide} \end{array} \right\}$ as in ikyagh, ikyegh, ikyôngo
- (b). g $\longrightarrow$ gʲ / $\left\{ \begin{array}{l} +\text{i- initial} \\ +\text{dorsal} \\ +\text{velar} \\ +\text{glide} \end{array} \right\}$ as in igyar, igyuve, igyongo
- (c). ŋ $\longrightarrow$ ŋʲ / $\left\{ \begin{array}{l} +\text{i- initial} \\ +\text{dorsal} \\ +\text{velar} \\ +\text{glide} \end{array} \right\}$ as in ingyaa, ingyur, ingyor

Dorsal palatalisation takes place in Tiv nouns whose initial syllable is the front high unrounded vowel /i/ followed by a syllable whose onset has a velar consonant + palatal glide /y/. Dorsal palatalisation in Tiv is secondary palatalisation in that, in the course production of the velar consonants /k/, /g/ and /ŋ/, it is the back portion of the tongue body, which is active during the consonantal gesture, while the front part of the tongue body is active in

performing the palatal trigger vocalic gesture. This explains why minimal overlap results in secondary palatalisation of velar consonants and why overlap of tongue body gestures does not automatically mean gestural blending. It is the degree of overlap that matters, minimal versus larger.

Finding and conclusion

The study finds that secondary palatalisation in Tiv nouns is both morphologically and phonologically conditioned. Morphologically induced secondary palatalisation in Tiv nouns occurs in the environments where the front high unrounded vowel /i/ is prefixed to the nominal base whose initial consonants is any of the following /p/, /b/, /t/, /d/, /k/, /kp/, /g/, /gb/, /f/, /v/, /m/, /mb/, /ŋ/, /h/ and /l/. The prefixation of /i/ to the nominal base further conditions the insertion of /j/ immediately after the target consonants in the second syllable. The consequence of the insertion of the /j/ is a change in the constriction location (CL) and constriction degree (CD) of the target consonants, as the tongue moves to the palatal region.

The study reveals that phonologically conditioned secondary palatalisation in Tiv nouns occurs in an environment where the physiological make up of a singular noun contains the front high unrounded vowel /i/ as initial syllable, accompanied by a syllable whose onset is any of the following consonants /p/, /b/, /t/, /d/, /k/, /kp/, /g/, /gb/, /f/, /v/, /m/, /mb/, /ŋ/, /h/ and /l, and the palatal glide /j/ as shown in example (2) without a change in its morphological make up.

The study discovers that apart from being morphologically and phonologically conditioned, secondary palatalisation in Tiv nouns is based on the following: (a) The presence of the front high unrounded /i/ as a v-shaped syllable in the physiological configuration of the noun as a pre-palatalisation clue. (b) The presence of /j/ which functions a palatalisation trigger immediately after the onset of the second syllable. (c) A change in the constriction location (CL) and constriction degree (CD) of the target consonants. (d) All target consonants are palatalised before the palatal glide/j/.

The study realises that secondary palatalisation in Tiv nouns is restricted to the onset of the second syllable as permitted by the Tiv phonotactics.

Recommendations

This study recommends that more research should be done to reveal other palatalisation patterns in Tiv. This recommendation is based on the fact that palatalisation in Tiv is not only secondary palatalisation, nor is it restricted only to nouns, and cannot be covered in a single study of this type.

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